

Appendix D

Non-equilibrium thermodynamics

D.1 Entropy production

When a system relaxes from a non-equilibrium situation, entropy is produced. Irreversible thermodynamics, as formulated by Onsager [55,56], establishes relations between the different contributions to the entropy production. As a starting point, we, therefore, need an expression for the entropy production. The “canonical” derivation for the expression of the entropy production is given in the book on *Non-equilibrium Thermodynamics* by De Groot and Mazur [57]. However, the derivation of the expression for entropy production in ref. [57], whilst complete, is a bit daunting. Here, we have opted for a different route: it is less complete, but it is quick, and it makes it easier to understand what is going on.

Let us first briefly look at the entropy generation that takes place if we bring two sub-systems (I and II) of a closed system into contact. The subsystems can exchange energy and particles. We ignore the fact that the medium may flow—and that is the price we have to pay for keeping things simple. In reality, viscous flow is an important non-equilibrium phenomenon, and the way to include it in the expression for the entropy production is described in ref. [57].

The change of the entropy, S of system I due to an infinitesimal exchange would be:

$$dS^I = \frac{1}{T^I} dU^I - \sum_i \frac{\mu_i^I}{T^I} dN_i^I. \quad (\text{D.1.1})$$

As the total system is closed, we have $dU^I = -dU^{II}$, and $dN_i^{II} = -dN_i^I$. Hence, the total change in entropy is

$$dS^{\text{total}} = dS^I + dS^{II} = \left[\frac{1}{T^I} - \frac{1}{T^{II}} \right] dU^I - \sum_i \left[\frac{\mu_i^I}{T^I} - \frac{\mu_i^{II}}{T^{II}} \right] dN_i^I. \quad (\text{D.1.2})$$

This equation shows that entropy is produced if there is an energy flux in a system with a gradient in $1/T$ or if there is a particle flux in a system with a gradient in μ_i/T . Now consider the case that we have a thin slab of material with width dx and cross-section A and that heat flows into the system at temperature

$T(x)$ and out of it at temperature $T(x + dx)$. We denote the amount of heat transported per unit of time by $\dot{Q}$. The heat flux is then $\mathbf{j}_q = \dot{Q}/A$.

Similarly, the number of particles of type i flowing through the system is $\dot{N}_i$ and the particles flux of species i is $\mathbf{j}_i = \dot{N}_i/A$. Note that we assume that the system as a whole is at rest and, therefore, the sum of all particle fluxes must vanish.¹

We can then write the rate of entropy production $\dot{S}$ as

$$\dot{S} = \left(\frac{\partial 1/T(x)}{\partial x} \right) dx A \mathbf{j}_q - \sum_i \left(\frac{\partial \mu_i(x)/T(x)}{\partial x} \right) dx A \mathbf{j}_i, \quad (\text{D.1.3})$$

or, in terms of σ , the entropy production per unit volume:

$$\sigma = \left(\frac{\partial 1/T(x)}{\partial x} \right) \mathbf{j}_q - \sum_i \left(\frac{\partial \mu_i(x)/T(x)}{\partial x} \right) \mathbf{j}_i. \quad (\text{D.1.4})$$

We can generalize this slightly by dropping the assumption that the transport is one-dimensional, and we can combine the gradient of the chemical potential with other forces $\mathbf{F}_i$ acting on species i , due to gradients in external potentials. We then obtain

$$\sigma = \mathbf{J}_q \cdot \nabla \frac{1}{T} - \frac{1}{T} \sum_i \mathbf{J}_i \cdot \left(T \nabla \frac{\mu_i}{T} - \mathbf{F}_i \right). \quad (\text{D.1.5})$$

For reasons to be discussed below, it is convenient to separate the gradient of μ_i/T into a part that depends on the temperature gradient and a part that does not. If the chemical potential depends on temperature and pressure, we can use

$$\left(\frac{\partial \beta \mu_i}{\partial \beta} \right)_{P, \{N_j\}} = h_i, \quad (\text{D.1.6})$$

where $\beta \equiv 1/k_B T$, and

$$\sigma = \left(\mathbf{J}_q - \sum_i \mathbf{J}_i h_i \right) \cdot \nabla \frac{1}{T} - \frac{1}{T} \sum_i \mathbf{J}_i \cdot [(\nabla \mu_i)_T - \mathbf{F}_i] \quad (\text{D.1.7})$$

and this allows us to define the ‘non-diffusive’, irreversible heat flow $\mathbf{J}'_q$

$$\mathbf{J}'_q \equiv \mathbf{J}_q - \sum_i \mathbf{J}_i h_i. \quad (\text{D.1.8})$$

¹ The most elegant choice for the particle fluxes would be the mass flux, in which case the condition that the sum of fluxes must vanish means that the center of mass is stationary. However, in practice, it makes little difference which fluxes we use. We will use number densities. The thing to bear in mind is that the definition of the chemical potential (e.g., per particle, per unit mass, or per mole) has to be consistent with the choice of the fluxes.

In what follows, we shall use the symbol $\mathbf{J}_h$ to denote the enthalpy flux:

$$\mathbf{J}_h \equiv \sum_i \mathbf{J}_i h_i. \quad (\text{D.1.9})$$

D.1.1 Enthalpy fluxes

Why is it important to subtract the enthalpy fluxes? First of all: as de Groot and Mazur state, heat fluxes in mixtures are not uniquely defined. The same holds for enthalpy fluxes. The value of the enthalpy flux depends on where we choose the zero of the energy of the particles. We could, for instance, include the $m_i c^2$ associated with the rest-mass of particle i . This is not as silly as it seems because if we are, for instance, pumping UF_6 or a similar nuclear fuel, the energy flux does take the rest of the mass into account.

The key point is that the “internal energy” choice makes a huge difference in what we mean by “transferring a particle at constant energy.” If we move particle i from reservoir 1 to reservoir 2, without moving energy, then removing particle i from reservoir 1 will lead to a large increase in the entropy (because the energy that used to be in the rest mass of particle i is now taken up by the bath of other particles), and conversely, the entropy of system 2 will decrease very substantially to compensate for the energy gain associated with the introduction of particle i . It is convenient to view this process *as if* we allow particle i to retain its enthalpy but then add an enthalpy flux from 2 to 1 that exactly compensates for this. Schematically, the problem of transferring a particle at constant energy is sketched in Fig. D.1.

In particular, we could consider the case where, in addition, there is a heat flux $\sum_i \mathbf{J}_i h_i$ from 1 to 2. This would correspond to the case that we do not force the particle to extract energy when it leaves 1 and to add energy when it arrives in 2. In this particular case, the contribution of diffusive enthalpy transport to the net ‘irreversible’ heat flow $\mathbf{J}'_q$ is zero. Importantly, if we thus allow particles to travel with their associated enthalpy, the entire problem of the reference state disappears. Of course, in reality, there still will be a heat flux, but this is the heat associated with thermal motion and intermolecular interactions.

To summarize: the rate of entropy production is given by

$$\sigma = \mathbf{J}'_q \cdot \nabla \frac{1}{T} - \frac{1}{T} \sum_i \mathbf{J}_i \cdot [(\nabla \mu_i)_T - \mathbf{F}_i] \quad (\text{D.1.10})$$

where the irreversible heat flow depends neither on the enthalpy carried by the particle fluxes $\mathbf{J}_i$.

D.2 Fluctuations

In the previous sections, we argued that the equilibrium state of a system corresponds to the state with the largest number of microscopic realizations. More-



FIGURE D.1 Transferring particles from system 1 at temperature T_1 to system 2 at temperature T_2 , without changing the energy of either system, requires that every particle leaves *all* energy (even kinetic) behind in 1 and acquires its new energy at the expense of system 2.

over, we argued that we could link this probabilistic picture to the experimental observation that the entropy of a closed system is at a maximum if we identify the entropy S with $k_B \ln \Omega$:

$$S = k_B \ln \Omega . \quad (\text{D.2.1})$$

We now assume that the entropy is a unique function of n linearly independent extensive variables, denoted by $\{A_1, A_2, \dots, A_n\}$. In equilibrium, the entropy of an isolated system must be at a maximum. If A_i denotes a non-conserved quantity (e.g., the degree of crystallinity), then the second law implies that, at equilibrium,

$$\frac{\partial S_t}{\partial A_i} = 0 , \quad (\text{D.2.2})$$

where S_t denotes the entropy of the entire system. However, if A_i denotes a conserved quantity, then it cannot change in a closed system. In that case, we can consider the entropy variation as an amount dA_i is transferred from subsystem

1 to subsystem 2. Then

$$\frac{\partial S_t}{\partial A_i^{(2)}} = \frac{\partial S_2}{\partial A_i^{(2)}} - \frac{\partial S_1}{\partial A_i^{(1)}} = 0, \quad (\text{D.2.3})$$

because $dA_i^{(2)} = -dA_i^{(1)}$. More generally, if we consider m subsystems, there are $m - 1$ independent variables $A_i^{(n)}$ if A is conserved, and m if A is not conserved. For the conserved quantities, Eq. (D.2.3) simply expresses the equality of T , P/T , or μ_i/T in the two subsystems. So, S does not vary to linear order in A_i . However, to quadratic order, S does vary with A_i .² Hence, to quadratic order, we can write

$$S = S_0 + \frac{1}{2} \sum_{i,j} \frac{\partial^2 S_t}{\partial A_i \partial A_j} \alpha_i \alpha_j, \quad (\text{D.2.4})$$

where $\alpha_i \equiv A_i - A_i^0$. To keep our notation aligned with De Groot and Mazur [57], we write

$$g_{ij} \equiv -\frac{\partial^2 S_t}{\partial A_i \partial A_j}. \quad (\text{D.2.5})$$

If the α_i are linearly independent, g_{ij} is a symmetric, positive definite matrix. The probability to find a system in a state away from (but close to) the most probable state and characterized by the variables $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$ is

$$P(\{\alpha_1, \alpha_2, \dots, \alpha_k\}) \sim \exp \left(-\frac{1}{2k_B} \sum_{i,j} g_{ij} \alpha_i \alpha_j \right). \quad (\text{D.2.6})$$

As the second law states that a system will evolve from a less probable to a more probable state, we can define the “driving force” that causes the system to return to its most probable state. With every variable α_i , there is an associated driving force X_i given by

$$X_i = -\frac{1}{2} \frac{\partial \sum_{i,j} g_{ij} \alpha_i \alpha_j}{\partial \alpha_i} = -\sum_j g_{ij} \alpha_j. \quad (\text{D.2.7})$$

To define the driving forces X_i , we made use of the assumption of *local* thermodynamic equilibrium, meaning that locally the relations between the entropy S and the basic extensive thermodynamic quantities U , V , and M_i (the mass of component i) are the same as during a quasi-static process. We do not know *a priori* how quickly a system returns to equilibrium. A separate set of constitutive

² We assume that S is an analytical function of the A_i . This may seem reasonable, but it need not always be true.

equations is needed to describe the relation between the driving forces X_j and the “fluxes” $J_i \equiv \dot{\alpha}_i$. We assume that, to lowest order, these relations are of the form:

$$J_i = \sum_j L_{ij} X_j . \quad (\text{D.2.8})$$

At this stage, we know nothing about the transport coefficients L_{ij} . Onsager [55] assumed that the law describing the rate of decay of a variable α_i to its equilibrium value (zero) is valid for arbitrarily small α_i and therefore also describes the rate at which spontaneous fluctuations around equilibrium decay to their average value. This “Onsager Regression Hypothesis” provides a link between the macroscopic transport coefficient and the microscopic dynamics of a system in equilibrium. The regression hypothesis can be viewed as a generalization of Einstein’s assumption that diffusive transport can be treated as a macroscopic manifestation of Brownian motion.

From Eq. (D.2.4) it is easy to derive an expression for the entropy production $\dot{S}$:

$$\dot{S} = - \sum_{i,j} g_{ij} \dot{\alpha}_i \alpha_j = \sum_i X_i \cdot \dot{\alpha}_i = \sum_{ij} L_{ij} X_i X_j . \quad (\text{D.2.9})$$

D.3 Onsager reciprocal relations

To make contact with the decay of fluctuations in equilibrium, we now show that the equilibrium fluctuations α_i are only correlated to the fluctuations in the *conjugate* force X_i

$$\langle \alpha_i X_j \rangle = -k_B \delta_{ij} \quad (\text{D.3.1})$$

where k_B is Boltzmann’s constant and δ_{ij} is the Kronecker delta. Equation (D.3.1) follows from the fact that

$$\begin{aligned} \langle \alpha_i X_j \rangle &= k_B \int d\{\alpha\} \alpha_i \frac{\partial P(\{\alpha\})}{\partial \alpha_j} \\ &= -k_B \int d\{\alpha\} \frac{\partial \alpha_i}{\partial \alpha_j} P(\{\alpha\}) \\ &= -k_B \delta_{ij} , \end{aligned} \quad (\text{D.3.2})$$

where $\{\alpha\}$ denotes the set $\{\alpha_1, \alpha_2, \dots, \alpha_k\}$. Using $J_i = \sum_j L_{ij} X_j$, it then follows that

$$\langle \alpha_j(t) J_i(t) \rangle = -k_B L_{ij} . \quad (\text{D.3.3})$$

Equation (D.3.3) allows us to derive the Onsager reciprocal relations. But before we do so, note that microscopically, Eq. (D.3.3) is a bit strange, because a fluctuation at time t will only result in a flux for $t > 0$. In fact, as α_j and J_j

have different time-reversal symmetries, the equal-time product actually vanishes. The quantity that is non-zero is $\langle \alpha_j(t) J_i(t + \epsilon) \rangle$. We will come back to this point later. For the time being, we continue with Eq. (D.3.3) and rewrite it as:

$$\int_0^\infty dt \langle \alpha_j(0) \dot{J}_i(t) \rangle = -k_B L_{ij} , \quad (\text{D.3.4})$$

where we have made use of the fact that $\langle \alpha_j(0) \dot{J}_i(0) \rangle$ vanishes. Next, using time invariance,

$$\int_0^\infty dt \langle \dot{\alpha}_j(0) J_i(t) \rangle = +k_B L_{ij} , \quad (\text{D.3.5})$$

and, using the fact that $J_j = \dot{\alpha}_j$:

$$\int_0^\infty dt \langle J_j(0) J_i(t) \rangle = k_B L_{ij} . \quad (\text{D.3.6})$$

The important thing is that (classically):

$$\langle J_j(0) J_i(t) \rangle = \langle J_i(t) J_j(0) \rangle = \langle J_i(0) J_j(-t) \rangle .$$

Here, and in what follows, we will limit the discussion to fluxes that all have the same time-reversal symmetry, in which case $\langle J_j(0) J_i(t) \rangle = \langle J_i(0) J_j(t) \rangle$, from which it then follows that

$$L_{ij} = L_{ji} . \quad (\text{D.3.7})$$